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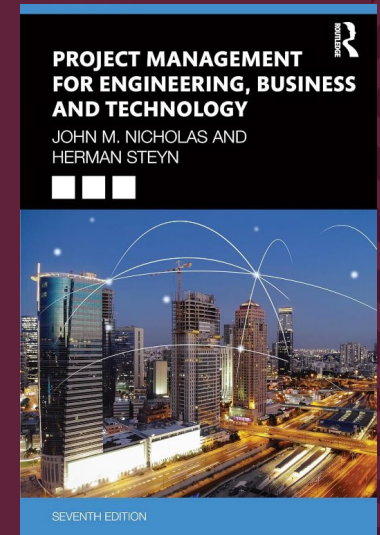
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CHAPTER 10: PROJECT QUALITY MANAGEMENT

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Adapted from slides
prepared by

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Herman Steyn*



Handbook



10.1 The Concept of Quality

10.2 Project Quality Management Process

10.3 Techniques for Quality Assurance in System Development

10.4 Techniques for Quality Control

Introduction



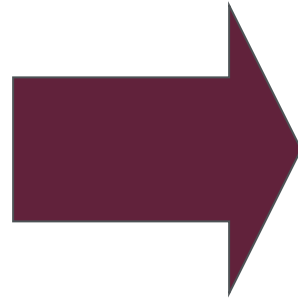
- Apart from meeting the client's expectation relating to **time** and **cost**, we need to satisfy the expectation around the product / system in order to achieve project success. Linked to the Scope aspect are therefore the issue of performance and **Quality**

10.1 The concept of Quality

1950s

Screening out defects

Inspection



Current

Culture

Processes

Including to
prevent defects &
failures

10.1 The concept of Quality



- In projects we are trying to meeting the specifications or requirements
 - Not more, or less, we are talking about fit for purpose
- A defect is a nonconformity – problem or misstate
 - To achieve quality, we need to identify nonconformities as soon as possible
 - The longer we wait, the bigger the impact will be on the project (Time & Cost)
- Total Quality also refer to time and cost dimensions
 - Zero defect might have a negative effect on the Time and Cost dimensions
 - Depend on the project / product / client

10.2 Project Quality Management Processes(1)



- Project Quality Management includes the management **processes** and **techniques** to lower risk of the product / system **not** meeting the requirements
 - This includes three processes:
 - Quality Planning, Quality Assurance, Quality Control
- Quality Planning
 - Guide future quality **activities**
 - Sets **requirements** and standards which should be met
 - Sets **actions** which should be taken to met requirements
 - Quality planning two aspects:
 - Establishing quality management **procedures and policies** (organisation)
 - Establishing quality plan as part of the **project** being executed

10.2 Project Quality Management Processes(2)



- Quality Planning
 - Cost of Quality
 - **Cost-benefit** analysis on **quality activities**
 - Prevention Cost, Appraisal Cost, Internal Failure cost, External Failure Cost
 - Project Quality Management Plan (PQMP)
 - Part of the Project Execution Plan
- Quality Assurance
 - Assure that the planned quality activities (according to the PQMP) are done
 - Happens on two levels:
 - Making sure that the current project are executed according to the PQMP
 - Making sure that we continuously improving in what we do now, for future projects
 - Providing **confidence** that we are doing everything we can to ensure the quality of the deliverables

10.2 Project Quality Management Processes(3)



- Quality Control
 - Ensuring the the quality assurance activities are done and that requirements and standards are being met
 - Include **planned** quality control activities, and **ad hoc** problem solving

10.3 Techniques for QA in System Development



1. Configuration Management
2. Design Reviews
3. Audits
4. Classification of Characteristics
5. FMEA
6. Modeling and Prototyping

10.4 Quality Assurance Techniques



- **Configuration Management**
 - Keeping track of iterations and versions of specifications, schematics, drawings, etc.)
 - Making sure we are working on the current relevant information
- **Design Reviews**
 - Making sure that the design requirements for the project are being met, and documented
- **Audits**
 - Verify management process comply
 - Verify technical processes were followed
- **Other**
 - Classification characteristics, Failure mode and effect analysis, Modelling and prototyping, System inspecting and testing, Document Inspection

10.4 Quality Control Techniques



- Final product inspection and acceptance testing
 - Verify the the product meet its specifications
- Tools of Quality Control
 - Run chart, Control chart, Pareto diagram, cause-and-effect diagram

Thank you
Enkosi
Dankie



Photo by Stefan Els